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How side effects can improve treatment efficacy: a randomized trial

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While treatment side effects may adversely impact patients, they could also potentially function as indicators for effective treatment. In this study, we investigated whether and how side effects can trigger positive treatment expectations and enhance treatment outcomes.

In this pre-registered trial (DRKS00026648), 77 healthy participants were made to believe that they will receive fentanyl nasal sprays before receiving thermal pain in a controlled experimental setting. However, nasal sprays did not contain fentanyl, rather they either contained capsaicin to induce a side effect (mild burning sensation) or saline (inert). After the first session, participants were randomized to two groups and underwent functional MRI. One group continued to believe that the nasal sprays could contain fentanyl while the other group was explicitly informed that no fentanyl was included. This allowed for the independent manipulation of the side effects and the expectation of pain relief.

Our results revealed that nasal sprays with a side effect lead to lower pain than inert nasal sprays without side effects. The influence of side effects on pain was dependent on individual beliefs about how side effects are related to treatment outcome, as well as on expectations about received treatment. Functional MRI data indicated an involvement of the descending pain modulatory system including the anterior cingulate cortex and the periaqueductal gray during pain after experiencing a nasal spray with side effects.

In summary, our data show that mild side effects can serve as a signal for effective treatment thereby influencing treatment expectations and outcomes, which is mediated by the descending pain modulatory system. Using these mechanisms in clinical practice could provide an efficient way to optimize treatment outcome. In addition, our results indicate an important confound in clinical trials, where a treatment (with potential side effects) is compared to placebo.

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Introduction

Common thinking in modern medicine posits that ideal treatments should have no side effects, because they can cause discomfort, suffering and treatment discontinuation. In addition, the mere expectation of side effects increases the likelihood of side effects.[1] Consequently, it has been suggested to not only minimize side effects, but also to carefully disclose information regarding potential side effects to avoid these nocebo effects.[2]

Here we challenge this view and ask whether some side effects could actually lead to better treatment outcomes. This idea is motivated by the observation that side effects themselves can contribute to treatment expectations.[3] Our hypothesis posits that even

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mild side effects can be indirect indicators of treatment potency (e.g. side effects are unavoidable with a powerful drug), which can lead to positive treatment expectations. These treatment expectations are the basis for non-specific therapeutic effects (i.e. placebo effect) that have substantial impact on treatment outcomes.[4-6]

Support for this hypothesis comes from research on active placebos, i.e. pharmacological agents that have a noticeable effect on the patient but not on the primary symptoms. A previous study on pain suggested that active placebos can indeed lead to larger placebo effects than inert placebos.[7] This idea also resonates with the observation that general practitioners prescribed more impure (i.e. active) placebos than inert placebos.[8]

The question of whether side effects can influence treatment outcome is also relevant for randomized clinical trials, as most studies compare active pharmacological interventions with inert placebos.[9] If the experience of side effects can indirectly boost treatment outcome, a clinical trial might overestimate the beneficial effect of active treatments that have an easily identifiable side effect profile.

To address these questions, we designed a multi-step experiment to investigate how side effects influence treatment efficacy on the psychological and neural level in a large sample of healthy volunteers using functional MRI (fMRI) (Fig. 1). The study was done in a controlled experimental placebo paradigm to exclude hidden pharmacological effects and isolate psychological and neural mechanisms, therefore it is not related to any drug-specific pharmacological effects and can be generalized to other treatments. According to our pre-registration (https://drks.de/search/de/trial/DRKS00026648), we hypothesized that side effects act as a cue for effective treatment and influence pain and placebo effects by augmenting expectation mechanisms. With respect to neuronal effects, we expected that side effects would recruit the descending pain modulatory system. Based on previous studies investigating descending pain modulation during placebo analgesia observing modulated coupling between the rostral anterior cingulate cortex (rACC) and the periaqueductal grey (PAG),[10,11] we hypothesized that the coupling between the rACC and the PAG would also be modulated during side effect induced pain relief.

Materials and methods

Participants

One hundred and four healthy participants were enrolled in this study. Nine participants were excluded due to problems with the nasal spray [the hospital pharmacy prepared a 1-week supply of nasal spray with an incorrect (too low) capsaicin dose], one due to a technical problem and two due to floor effects of applied pain [mean ratings <10 on a visual analogue scale (VAS) from 1 to 100]. Fifteen participants were excluded after performing the pre-registered manipulation check (see 'Manipulation check' section). Seventy-seven participants were included in the data analysis [Expectancy group: n = 42, age: 24.3 ± 3.8 (18–38), 12 males; No-Expectancy group: n = 35, age: 24.8 ± 5.4 (18–43), 14 males]. Sixty-seven participants were included in the data analysis for the follow-up experiment [Expectancy group: n = 36, age: 24.5 ± 3.9 (20–38), 11 males; No-Expectancy group: n = 31, age: 24.9 ± 5.6 (18–43), 12 males]. Participants were confirmed to be healthy through an in-person interview with a medical doctor during the initial visit. Inclusion criteria included age between 18–45 and body mass index (BMI) 18–30. Exclusion criteria included somatic or psychiatric conditions, regular intake of medications (excluding hormonal contraception, allergy medications, thyroid medication), recent (<24 h) intake of pain medication, drug abuse, pregnancy or breast feeding, skin diseases or skin damage on the forearms, nasal diseases or mucosal damage, allergy against pain medications, as well as any conditions that might affect MRI safety. Participants performed a drug test during the initial visit and were ineligible to participate in case of a positive result [amphetamines, benzodiazepines, buprenorphine, cocaine, morphine, methadone metabolites and THC (marijuana)]. Participants were recruited using flyers and internet advertisements and were compensated for their participation. Data collection was performed between October 2021 and June 2022 at the University Medical Center Hamburg-Eppendorf. All participants gave written consent according to the Declaration of Helsinki and the study was approved by the Ethics Committee of the Hamburg Medical Association (PV7226).

Experimental paradigm

The experiment consisted of three visits. During the initial visit, participants were informed about the research, checked for eligibility and signed consent forms. Participants underwent a medical instruction session where they received information that the aim of the study was to investigate the neural processes associated with fentanyl nasal spray, a powerful analgesic drug used in the treatment of cancer pain. They were informed about the medication's purpose and potential side effects of fentanyl, which included a burning sensation in the nose (Supplementary material). Afterwards, basic vital information (height, weight, blood pressure) was assessed and a drug screening was performed. Participants then completed a series of questionnaires, including questions about their belief that side effects indicate a more potent treatment.

During the second visit the primary experiment was performed: critical instructions were repeated and a heat pain sensitivity assessment was performed. Heat pain stimuli were applied using a thermode (PATHWAY System). Participants were then instructed on how to use a VAS to rate their pain (ranging from 0 = no pain to 100 = maximum tolerable pain). Finally, participants rated the intensity of a series of painful heat stimuli (n = 12) and a first order polynomial (line) was fitted to the data. Using these data, we established temperature levels corresponding to individual pain at VAS 40, 60 and 70.

During the experiment, participants believed that they receive three nasal sprays in three experimental runs, each with a 50% chance of containing fentanyl, and that this procedure will be repeated in the MRI scanner as well as 7 ± 1 days later (Fig. 1). Unbeknownst to the participants, none of the nasal sprays contained any fentanyl. However, one of the three nasal sprays (applied second or last) contained a small dose of capsaicin (0.15 μg/puff), which causes a burning sensation in the nose. All nasal sprays had differently coloured labels indicating the run number to facilitate recognition across sessions for participants. After the application of each nasal spray, participants had to rate several items regarding potential side effects on a four-point scale ('none at all', 'minimally', 'a bit', 'clearly'), including experiencing a burning sensation in the nose. Before each run, the thermode was moved to avoid sensitization or habituation and three warm-up stimuli were applied (20 s). Each run consisted of 24 trials and each trial consisted of anticipation (1.5 s–2.5 s), pain stimulation (6.5 s with 4 s pain plateau), VAS pain rating (8 s) and a variable inter-trial interval (7–9 s).

The difference between the runs was the potential side effect experience (capsaicin or no capsaicin) and the applied temperature

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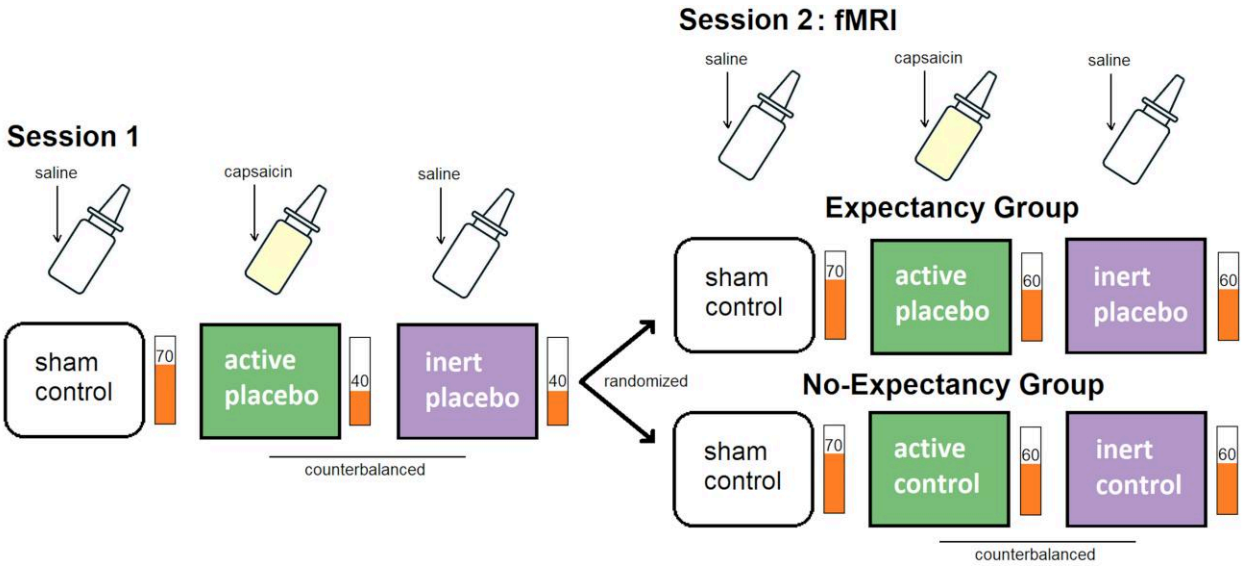


Figure 1 Experimental design. In the first session, participants underwent three separate runs, each involving the administration of a different nasal spray. Participants were informed that each spray could potentially contain the potent painkiller, fentanyl (uniform instruction for all three runs). Unbeknownst to the participants, none of the nasal sprays actually contained active pain medication. However, one of the sprays did contain a small dose of capsaicin, inducing a mild but perceptible side effect—a burning-like sensation in the nose. Following the nasal spray application, a series of thermal pain stimuli was administered. These stimuli were previously calibrated to levels of 40 (placebo conditions in the first session), 60 (placebo/control conditions in the second session) or 70 (sham control) on a VAS ranging from 0 to 100. Participants then rated their pain on the visual analogue scale (VAS). The temperature reduction in the placebo conditions was employed to simulate a treatment benefit. In a subsequent functional MRI (fMRI) session, a similar procedure was repeated to investigate the neural mechanisms underlying how side effects can modulate treatment efficacy. To assess the role of expectation of treatment benefit, the sample was randomized into two groups before the second session. The Expectancy group believed that the experiment would be repeated and continued to expect that any pain reduction was due to the pain-relieving fentanyl nasal spray. In contrast, the No-Expectancy group was explicitly informed that no fentanyl was present in any nasal spray and that differences in pain were solely due to variations in applied temperature. Therefore, side effects and expectation of treatment benefit were manipulated independently, enabling an examination not only of their individual roles but also of the interaction between both factors.

(VAS 40, 60 or 70; Fig. 1). The first nasal spray always corresponded to the sham control condition. During this condition, the nasal spray did not cause any side effects (saline, no capsaicin) and pain corresponded to VAS 70. During the following two runs, participants received reduced pain corresponding to VAS 40 with either a saline nasal spray with capsaicin (active placebo) or without capsaicin (inert placebo). Stimulation temperature was reduced in both placebo conditions to mimic a treatment benefit compared to the previous sham control run. Both runs were the conditions of interest and the order was randomized across participants and concealed to participants and experimenters. Afterwards, participants answered questions regarding their expectation of what they received in the previous runs ('believed no-treatment', 'no belief', 'believed treatment') and on how sure they were about their answer (on a scale from 0—'very uncertain' to 10—'very certain').

The second session was performed on the same day as the first session. For the second session, volunteers were randomized into an 'Expectancy' and a 'No-Expectancy' group. Randomization of group and condition order was performed at the start of data collection using MATLAB's randperm function. The No-Expectancy group was debriefed to eliminate their expectation of pain relief, similar to previous experiments.[12] They were told that no fentanyl was present in any of the nasal sprays and that the pain was surreptitiously reduced during the second and third run. The Expectancy group was not informed and simply received the instruction that the same procedure will be repeated in the MRI scanner. During MRI scanning, participants completed the same paradigm as before, with the exception that during the second and third run they received pain corresponding to VAS 60. VAS 60 was used as a compromise between further reinforcing the placebo expectations in the Expectancy group and incorrectly inducing strong pain relief expectation in the No-Expectancy group. Participants were taken out of the MRI scanner between runs and moved into a seated position to apply the nasal sprays, so that they could see that the same nasal sprays were applied. All participants were invited for a 7 ± 1 day follow-up experiment; however, some participants declined the follow-up experiment or were unable to attend. During the follow-up visit, participants completed the same paradigm as before, with pain corresponding to VAS 60 in the second and third run (Supplementary Fig. 4). At the end of the experiment, the Expectancy group was also debriefed. To account for the deceptive component, participants were explicitly asked whether they would like to withdraw their data at this stage. However, none of the participants withdrew their approval.

Manipulation check

After the experiment, a structured interview was performed to determine how strongly participants believed that there was a possibility for them to receive fentanyl. The interview asked participants to describe their thoughts throughout the different stages of the experiment as well as directly inquired whether they suspected anything. After the structured interview, two separate scientists (T.F. and L.S.) read participant's statements independently and rated if the cover story was believed successfully. In case of disagreement, a third scientist (C.B.) read the statements and gave the deciding vote. This was done for all participants in both groups on a participant by participant basis to avoid inducing group differences.

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Data acquisition

Stimulus presentation and behavioural data collection was carried out using Psychtoolbox in MATLAB (MathWorks, Natick, MA, USA). Functional MRI data were acquired on a 3 T magnetic resonance system (Magnetom Prisma, Siemens) equipped with a 64-channel head coil. To measure blood oxygen level-dependent (BOLD) responses, a T_2^*-weighted 2D multiband gradient echo planar imaging sequence was used (repetition time: 1.9 s; echo time: 26 ms; field of view: 225 mm^2; multiband acceleration: 3). Each volume consisted of 78 transversal slices with a voxel size of 1.5×1.5×1.5 mm^3. High resolution anatomical T_1-scans were acquired using an MPRAGE sequence with a voxel size of 1×1×1 mm^3.

Behavioural data analysis

Behavioural data analysis was performed using SPSS 27 (IBM, Armonk, USA). Pain rating data and experienced side effects were analysed using repeated measures 2×2 ANOVAs with condition (active versus inert, within-subjects factor) and group (Expectancy versus No-Expectancy group, between-subjects factor) as predictors and pain ratings or side effect ratings as dependent variables, separate for each phase. The data of the sham condition were not used in any statistical analysis. For within-subject contrasts, paired *t*-test and Wilcoxon signed-ranks test were used, depending on scale and normal distribution. Effect sizes are provided as partial η^2 for ANOVAs, Cohen's *d* for *t*-tests and Cohen's *r* for Wilcoxon singed-ranks test. For the mediation analysis, differences (active – inert) were calculated for side effects, treatment expectations (with believed treatment = 2, no belief = 1, believed no-treatment = 0) and pain rating. The belief of how side effects influence treatment was assessed with their agreement with the statement: 'stronger treatments have more side effects' [five-point scale with end points labelled: completely agree to completely disagree; mean ± standard deviation (range): 2.45 ± 0.8 (1–5); normally distributed]. This was asked during the first visit within the questions of the Beliefs about Medicines Questionnaire (BMQ),[13] a questionnaire about cognitive representations of medications, to avoid pointing the participants towards our research question. Mediation analysis was then performed with PROCESS in SPSS (Model 8: moderated mediation).[14] Bootstrap with 5000 samples was used to test for significance and to account for non-normal distribution. All effects were considered significant at $P < 0.05$ (two-tailed).

Functional MRI data analysis

Functional MRI data preprocessing and statistical analyses were performed using SPM12 (Wellcome Department of Imaging Neuroscience, London, UK). Data preprocessing consisted of slice timing, motion correction and co-registration of the functional images to the T_1 anatomical scan. Finally, the images were spatially normalized using DARTEL (based on the IXI555 template from the CAT12 toolbox, http://dbm.neuro.uni-jena.de/cat/) and smoothed using a 6-mm full-width at half-maximum (FWHM) isotropic Gaussian kernel.

We performed a first level analysis using a general linear model in SPM12. Each regressor was modelled by boxcar functions convolved with a canonical haemodynamic response function (HRF). For each run, we included regressors for cue (1.5–2.5 s), pain (4 s) and pain rating (8 s). T-contrasts of interest were then calculated. T-tests between conditions of interest were used to test for significance. We employed a region of interest (ROI) approach according to our pre-registration and selected ROIs based on previous meta-analyses: for the interaction contrast [($active_{exp} > inert_{exp}$) >/< ($active_{no\text{-}exp} > inert_{no\text{-}exp}$)], we defined the following regions: rACC (0 36 6 mm; 20 mm diameter),[15] dorsolateral prefrontal cortex (DLPFC) (±39 23 33 mm; 20 mm diameter).[15] For the placebo contrast without side effects ($inert_{exp}$ >/< $inert_{no\text{-}exp}$), the ROIs comprised the rACC,[15] the DLPFC,[15] the anterior-mid insula (±39 −6 −4 mm; 16 mm diameter),[15] the secondary somatosensory cortex (S2) (±47 −16 13 mm; 16 mm diameter)[15] and the PAG (0 −29 −12 mm; 8 mm diameter).[16] For all lateral regions, two ROIs for the left and right hemisphere were defined. For medial regions (rACC and PAG), a single ROI (at $x = 0$ mm) was used. For each contrast of interest, all ROIs were combined into a single mask to avoid inflation of type-I error. Results were considered significant at $P < 0.05$ family-wise error (FWE) corrected.

To test for a modulation of coupling between the rACC and the PAG, we performed a Psycho-Physiological Interaction (PPI) analysis.[17] We extracted the time series within a 3 mm sphere around the peak activation of the rACC from the interaction contrast (−6 33 −1.5) during the pain phase. Then we calculated the PPI interaction term as the time series multiplied by the psychological predictor (pain versus no pain). All three regressors were subsequently included in a new first level analysis. After model estimation, *t*-contrasts of interest were calculated for the PPI. Correction for multiple comparisons for this analysis was performed in the PAG ROI, as we only specified this target in our pre-registration.[16]

For illustration purposes, all statistical maps use a significance threshold of $P < 0.005$ uncorrected and were overlaid on the mean structural image of all participants. All activations are reported using *x*, *y*, *z* coordinates in MNI (Montreal Neurological Institute) standard space.

Results

Side effects lead to improved pain relief and are mediated by treatment expectations

Pain ratings and side effects were analysed using a two condition (active versus inert) by two group (Expectancy versus No-Expectancy) ANOVA. For the first session, we established that participants experienced more side effects after active placebo [main effect of condition; $P < 0.001$; $F(1,75) = 1138$; $\eta_p^2 = 0.94$; Fig. 2A]. To test for pre-randomization baseline group differences, we also investigated the main effect of group and the interaction. As expected, no interaction [$P = 0.65$; $F(1,75) = 0.21$; $\eta_p^2 < 0.01$] or main effect of group [$P = 0.89$; $F(1,75) = 0.02$; $\eta_p^2 < 0.01$] was observed for side effects. While all participants expected pain relief from the nasal sprays, we observed that pain ratings were lower after active placebo as compared to inert placebo [main effect of condition; $P = 0.002$; $F(1,75) = 10.7$; $\eta_p^2 = 0.15$; VAS 26.2 ± 2.0 versus 30.2 ± 2.0] (Fig. 2B and Supplementary Fig. 1). As expected, there was also no interaction [$P = 0.60$; $F(1,75) = 0.27$; $\eta_p^2 < 0.01$] and no main effect of group [$P = 0.33$; $F(1,75) = 0.95$; $\eta_p^2 = 0.01$] for pain ratings. This clearly shows that the experience of side effects caused a pronounced pain relief.

When asked to rate whether they believe that a nasal spray contained fentanyl or not (Fig. 2C), most participants believed that the active placebo condition contained fentanyl (94.8%). Contrary to this, the inert placebo condition was more ambiguous and only a minority of participants believed that it contained fentanyl (16.9%). The distribution was significantly different between the active and inert condition ($P < 0.001$; $z = -7.2$; $r = -0.82$; Wilcoxon signed-ranks test). Consistently, the confidence in their rating was larger in the active placebo condition compared to the inert placebo condition [$P < 0.001$; $T(76) = 5.6$; $d = 0.64$; paired *t*-test; Supplementary Fig. 2].

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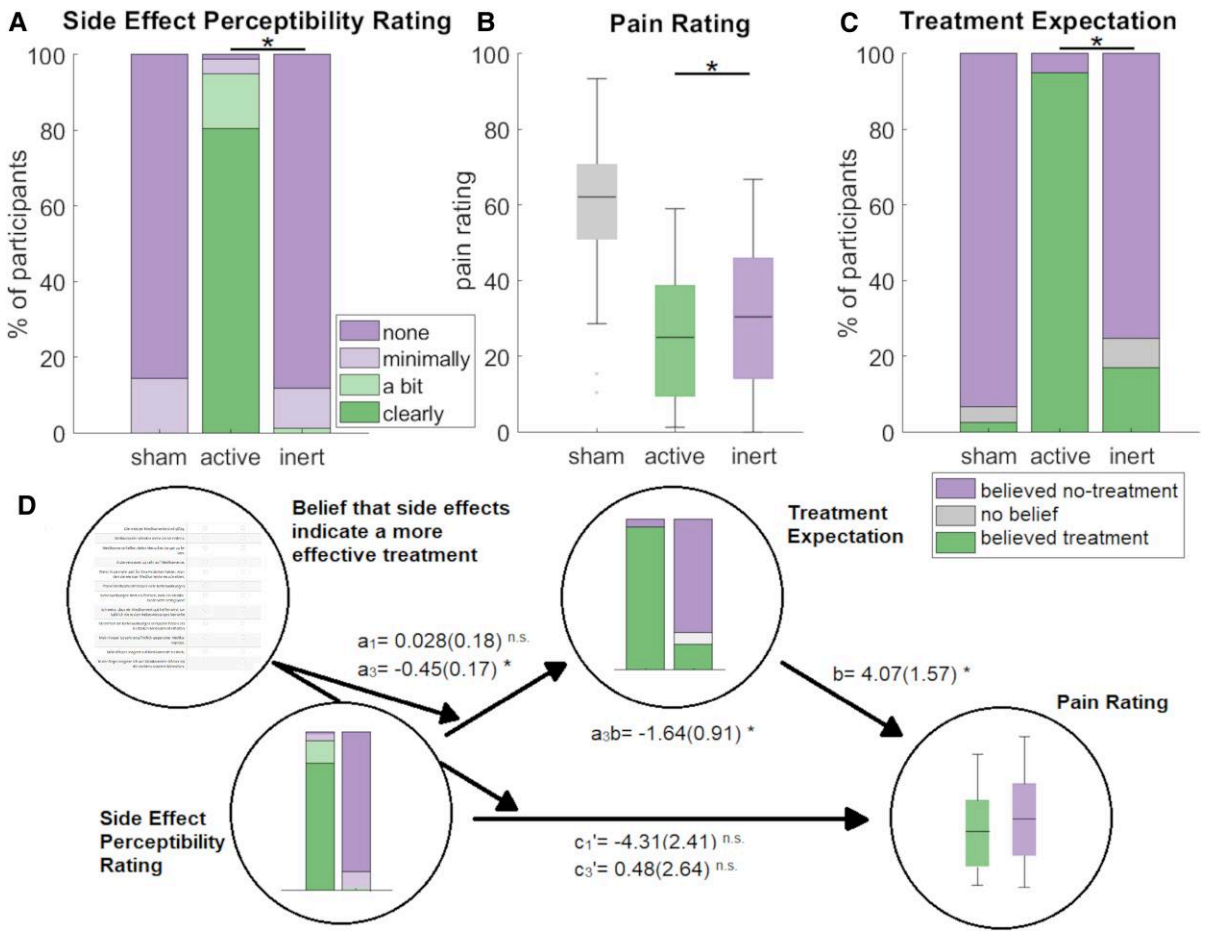


Figure 2 Session 1 results. Side effects lead to larger pain relief and are mediated by treatment expectations. (**A**) Participants reported more side effects after receiving an active placebo compared to inert placebo [main effect of condition: $P < 0.001$; $F(1,75) = 1138$; $\eta_p^2 = 0.94$; 2.74 ± 0.59 versus 0.13 ± 0.38]. (**B**) Pain ratings were lower after receiving an active placebo compared to inert placebo [main effect of condition: $P = 0.002$; $F(1,75) = 10.7$; $\eta_p^2 = 0.15$; VAS 26.2 ± 2.0 versus 30.2 ± 2.0], indicating that the experience of side effects influenced pain relief. (**C**) The majority of participants rated the sham condition as no treatment (no treatment: 93.5%; no belief: 3.9%; fentanyl: 2.6%). Almost all participants believed the active placebo contained fentanyl (fentanyl: 94.8%; no treatment: 5.2%). The inert placebo condition was the most ambiguous: although participants experienced pain relief compared to the sham condition, >83% did not believe that the nasal spray contained any medication (no treatment: 75.3%; fentanyl: 16.9%; no belief: 7.8%). All conditions were rated significantly differently from each other (all comparisons $P < 0.001$). (**D**) The belief that side effects indicate a more effective treatment moderated the relationship between experienced side effects and expectations. The expectation about the received treatment mediated the relationship between the experienced side effects and pain relief [indirect path: −1.64(0.91); 95% CI: (−3.98 −0.49)]. Box plots indicate median, interquartile interval, minimum and maximum. CI = confidence interval; n.s. = not significant; VAS = visual analogue scale.

In a next step, we tested whether the effect of side effects depended on the participants' expectations of having received fentanyl or no treatment, as well as on their belief that side effects indicate a more potent treatment (pre-registration). We therefore conducted a moderated mediation analysis to test whether their belief that side effects indicate a more potent treatment (W) moderated the effect of side effects (X) on treatment expectation (M) and if their expectation about having received fentanyl or no-treatment (M) mediated the relationship between experienced side effects (X) and pain relief (Y). We observed a significant moderation effect of beliefs that side effects indicate a more potent treatment on the association between side effects and treatment expectation [$a_{3(XW \rightarrow M)} = -0.45(0.17)$; 95% confidence interval (CI): (−0.82, −0.19)] with no main effect of side effects [$a_{1(X \rightarrow M)} = 0.028(0.18)$; 95% CI: (−0.31, 0.39)]. Additionally, treatment expectations had a significant effect on pain relief [$b_{(M \rightarrow Y)} = 4.07(1.57)$; 95% CI: (1.29, 7.51)]. Both direct effects [$c_1'_{(X \rightarrow Y)} = -4.31(2.41)$; 95% CI: (−9.24, 0.27); $c_3'_{(XW \rightarrow Y)} = 0.48(2.64)$; 95% CI: (−5.68, 4.73)] were not significant after mediation. Overall, we observed a moderated mediation [indirect path $a_3b = -1.64(0.91)$; 95% CI: (−3.98 −0.49); Fig. 2D], indicating that side effects did not directly influence pain relief, but that side effects, depending on their belief about how side effects affect treatments, influence their expectations about their treatment and that these, in turn, influenced pain relief.

Side effects lead to improved pain relief and are associated with increased rACC-PAG coupling

During the following MRI session, participants were randomly allocated into an Expectancy group that continued to believe that pain decreases were due to fentanyl, and a No-Expectancy group that was explicitly debriefed that no fentanyl was present. We observed a significant interaction between expectation and side effects: the Expectancy group continued to show lower pain ratings after active placebo compared to inert placebo, while the No-Expectancy group did not show this difference any more [interaction: $P = 0.04$; $F(1,75) = 4.3$; $\eta_p^2 = 0.054$] (Fig. 3A and Supplementary Fig. 1). We did not observe a main effect of condition [$P = 0.31$; $F(1,75) = 1.0$; $\eta_p^2 <$

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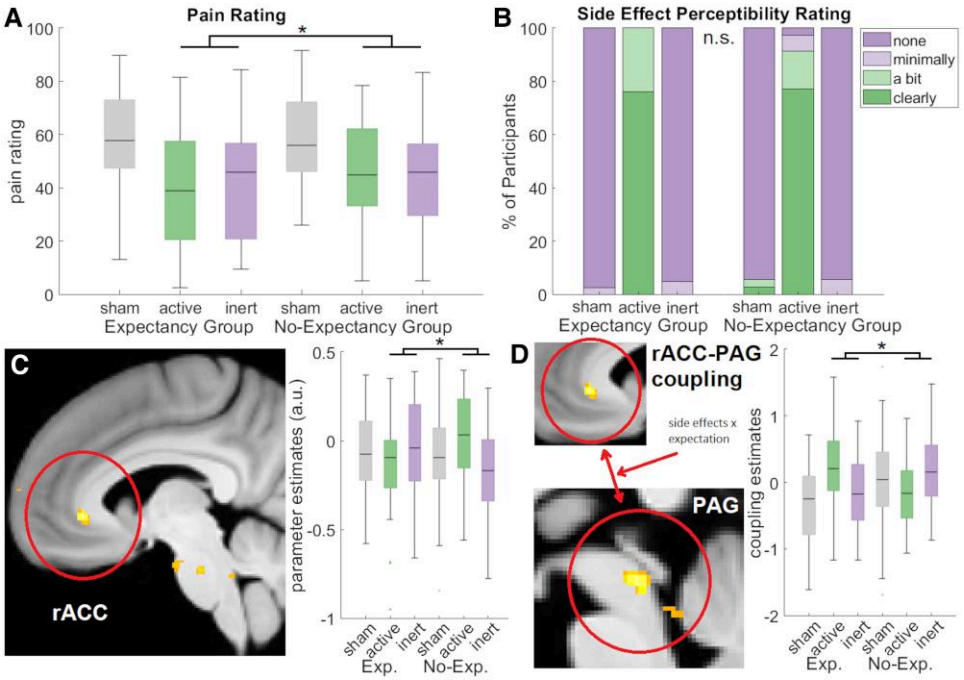


Figure 3 Session 2 fMRI results. Side effects lead to larger pain relief and are associated with increased rACC-PAG coupling. (**A**) The Expectancy group continued to show lower pain ratings after active placebo compared to inert placebo, while the No-Expectancy group did not show this difference any more [interaction: $P = 0.04$; $F(1,75) = 4.3$; $\eta_p^2 = 0.054$; VAS 38.7 ± 3.4 versus 43.0 ± 3.2 and VAS 45.0 ± 3.3 versus 43.5 ± 3.3]. (**B**) Side effects continued to be experienced after active placebo with no difference between groups [main effect of condition: $P < 0.001$; $F(1,75) = 1305$; $\eta_p^2 = 0.95$; 2.71 ± 0.58 versus 0.05 ± 0.22]. (**C**) We observed reduced BOLD signal during active placebo in the rACC {$T = 4.5$; $P = 0.003$; [($active_{exp} > inert_{exp}$) < ($active_{no\text{-}exp} > inert_{no\text{-}exp}$)]}. (**D**) rACC-PAG coupling was increased during active placebo {$T = 4.37$; $P < 0.001$; [($active_{exp} > inert_{exp}$) > ($active_{no\text{-}exp} > inert_{no\text{-}exp}$)]}, indicating a stronger recruitment of the descending pain modulatory system. Box plots indicate median, interquartile interval, minimum and maximum. fMRI = functional MRI; PAG = periaqueductal gray; rACC = rostral anterior cingulate cortex; VAS = visual analogue scale.

0.01] and no main effect of group [$P = 0.45$; $F(1,75) = 0.57$; $\eta_p^2 = 0.01$]. Investigating both groups separately, we observed a marginal significant effect in the Expectancy group [$P = 0.05$; $T(41) = 1.9$; $d = 0.30$] and no significant effect in the No-Expectancy group [$P = 0.33$; $T(34) = -0.98$; $d = -0.16$]. Side effects continued to be experienced after active placebo [main effect of condition: $P < 0.001$; $F(1,75) = 1305$; $\eta_p^2 = 0.95$] (Fig. 3B and Supplementary Fig. 3) with no group difference [$P = 0.44$; $F(1,75) = 0.47$; $\eta_p^2 < 0.01$] or interaction [$P = 0.50$; $F(1,75) = 0.60$; $\eta_p^2 < 0.01$]. This further supports the observation that pain relief is larger after experiencing a side effect and shows that this effect is modulated by the expectation of pain relief by a treatment, thereby further supporting that side effects lead to larger pain relief when treatment benefit is expected.

During this session, to investigate the neural basis of how side effects interact with treatment expectations, we used fMRI to test for BOLD differences during pain stimulation. We expected that increased pain relief due to side effects might be mediated by a stronger activation of the descending pain modulatory system. Based on previous studies,[10,11,18] we therefore pre-registered the hypothesis that the side effect by expectation interaction will be associated with a modulation of the rACC, the DLPFC, as well as a modulation of the coupling between the rACC and the PAG. Testing for the interaction, we observed a modulation of the rACC {$T = 4.5$; $P = 0.003$; peak: (−6 33 −1.5); contrast: [($active_{exp} > inert_{exp}$) < ($active_{no\text{-}exp} > inert_{no\text{-}exp}$)]} (Fig. 3C), indicating a reduced BOLD signal during active placebo in the Expectancy group, similar to previous work.[11] We did not observe a modulation of the DLPFC. To test for 'classic' placebo effects (i.e. in the absence of side effects), we also tested the contrast between the inert condition in the Expectancy group and the inert condition in the No-Expectancy group ($inert_{exp}$ >/< $inert_{no\text{-}exp}$). No brain region within the mask of interest reached statistical significance (rACC, DLPFC, anterior-mid insula, S2, PAG) (Supplementary Table 1). Finally, we tested if the coupling between the rACC and the PAG is increased during active versus inert placebo in the Expectancy group compared to the No-Expectancy group. We observed a significantly increased coupling {$T = 4.37$; $P < 0.001$; peak: (0 −30 −12); contrast: [($active_{exp} > inert_{exp}$) > ($active_{no\text{-}exp} > inert_{no\text{-}exp}$)]} (Fig. 3D), indicating a stronger activation of the descending pain modulatory network during active placebo.[19]

We also reinvited participants to repeat the behavioural part of the experiment 7 ± 1 days later (Supplementary Fig. 4). As in the first session, we again observed a main effect of condition, with lower pain ratings for the active placebo as compared to the inert placebos [$P = 0.004$; $F(1,65) = 9.0$; $\eta_p^2 = 0.12$]. However, the group difference was not significant [$P = 0.15$; $F(1,65) = 2.03$; $\eta_p^2 = 0.03$] and there was no interaction effect [$P = 0.93$; $F(1,65) < 0.01$; $\eta_p^2 < 0.01$] anymore. The difference between the active and inert condition was significant in the Expectancy group [$P = 0.03$; $T(35) = 2.2$; $d = 0.37$] and marginally significant in the No-Expectancy group [$P = 0.05$; $T(30) = 2.0$; $d = 0.37$]. This shows that after 7 ± 1 days, without any further expectation manipulation, active placebo continued to lead to larger pain relief. Although expectancy was not directly measured, this could potentially indicate that the debriefed No-Expectancy group re-established their expectation of treatment benefit. Further exploratory analyses can be found in Supplementary Figs 5 and 6.

Discussion

In this study, we show that side effects can improve pain relief through the amplification of treatment expectations. We also

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show that expectations about treatments mediate the effect of side effects on pain relief and that this is dependent on participants' individual beliefs of how side effects are related to treatments. Additionally, our fMRI results indicate that pain relief induced by active placebos involve a recruitment of the descending pain modulatory system.

Our data show that side effects do not directly act on the pain experience, but are mediated by treatment expectations, and are dependent on beliefs of how side effects are related to treatment effectiveness. This is in line with placebo research that shows that contextual factors primarily influence expectations that then in turn influence treatment outcome.[12] Therefore, if these mediating beliefs can be changed, treatment providers can influence how side effects affect treatment outcome.

We observed a decreased BOLD signal in the rACC. This seems to be at odds with the notion of the rACC being a pain modulatory region. Although many studies have reported higher rACC BOLD signal for placebo compared to control, several studies have also shown decreased rACC activity.[18] Importantly, a previous study[11] observed both rACC activation and rACC deactivation. The latter region was modulated by naloxone, suggesting a causal involvement of this region in pain modulation. The peak response in the rACC in the latter study is very close (<8 mm) to our observed response in the rACC and shows a similar activation pattern. Thus, our observed pattern of signal changes in the rACC is not unexpected, and more importantly linked to a brain region that has been shown to be causally involved in pain modulation.

Additionally, we observed a stronger rACC-PAG coupling when side effects boost pain relief. Previous research with fMRI,[20] or molecular imaging[21] has implicated this pathway in treatment expectation induced pain relief and descending pain modulation. Furthermore, rACC-PAG connectivity correlates with pain relief[11] and blocking μ-opioid receptors with naloxone can abolish both rACC-PAG coupling and expectation-induced pain relief.[11] Therefore, our results indicate that the increased expectation of treatment benefits that arise from the experience of side effects is at least in part due to the recruitment of the descending pain modulatory system.

Figures 3C and D show a non-significant difference between the active and inert condition in the No-Expectancy group as part of a crossed interaction. This might be related to other mechanisms that could have an influence on our observations, e.g. negative aspects of side effects might have influenced pain and rACC BOLD signal in the active control condition of the No-Expectancy group, possibly mediated by a similar mechanism that has been shown to increase pain by negative mood.[22] Future work may be needed to more fully understand this finding.

In contrast to our hypothesis, we did not observe a modulation of the DLPFC. Although the DLPFC is implicated in placebo analgesia in several studies,[18] our non-significant result with respect to the DLPFC is in line with a recent meta-analysis with individual participant data,[23] that could not confirm a significant effect in the DLPFC, possibly related to a large variability of the effect across individuals.

One additional aspect to consider is whether our effects are the result of conditioned pain modulation (CPM).[24] In CPM, a tonic painful stimulus (i.e. nasal capsaicin) renders a phasic painful stimulus (i.e. thermal pain) at a different body site less painful. However, CPM is unlikely to be relevant for our findings for two reasons: the very small dose of capsaicin inducing a mild burning sensation, as would be expected to occur as a normal treatment side effect, is unlikely to be sufficiently intense to induce CPM, as previous studies show that a high intensity tonic stimulus is required for CPM.[25] More importantly, our observed interaction during Session 2 (Fig. 3A) cannot be explained by CPM as in both groups the burning sensation in the nose is identical.

Despite experiencing pain relief during the inert condition compared to the initial sham condition, only a minority of participants believed that it contained the treatment (16.9%). Together with our data from the participant's responses about how side effects relate to treatments, this supports the notion that for many people, effective treatment is inherently associated with side effects. Although speculative, this could also be a potential explanation for the re-emergence of higher pain relief during Session 3 after 7 ± 1 days. In contrast to Session 2, which was conducted directly after debriefing, after 7 ± 1 days and without any further expectancy manipulation, the side effect again led to a large placebo effect, indicating a 're-established' belief of treatment efficacy through side effects.

It is also important to note that in our experiment, the side effect was designed to be mild and benign and therefore optimal for inducing positive treatment expectations. It is likely that an optimal window of side effect intensity exists, with too weak and unspecific side effects unable to trigger positive expectations, while too strong and harmful side effects might even lead to negative expectations.

Our study has also two major clinical implications. Using strategies to maximize positive treatment expectancies (i.e. placebo effects) in clinical practice can significantly improve many treatment outcomes.[4,5,26] Our data suggest that mild and benign side effects do not necessarily have to be harmful to patients and could potentially even resemble an overall benefit for treatment outcome. One could even think of artificially changing the formulation of established drugs to include mild side effects to increase treatment expectations. However, while research on open label placebo[27] provides the idea that this effect could be used without deception, great care would be necessary to avoid any unintended harm.

A simpler strategy to increase treatment benefit may lie in the framing of side effects.[28] Current expert consensus on how to inform patients about side effects emphasize the prevention of nocebo effects.[29] Studies show that optimized communication, such as positive framing of side effects, can indeed reduce nocebo effects.[2,30] Our mediation analysis shows that the effect of side effects on pain relief depends on beliefs that side effects are a sign of treatment effectiveness. Therefore, optimized framing of side effects as a cue that the treatment is acting and healing might not only reduce nocebo effects but also increase positive treatment expectations and placebo effects, which often lead to more beneficial treatment outcomes.[4,5,31] Our data also support the idea that this information can be conveyed without inducing additional nocebo effects (Supplementary Fig. 3). Therefore, our study shows the psychological and neural mechanism that could be used to achieve this improvement in framing of side effects, however, it would be important to replicate the effect in a clinical sample.

Our study also has important implications for the interpretation of placebo controlled randomized clinical trials. Methodological standards, such as random treatment allocation, double-blinding of patient and providers, and others have been established to achieve the assumption that non-specific factors (i.e. placebo effects) are additive between the different experimental conditions and therefore the treatment effect can be isolated.[32] As our treatment allocation instruction was probabilistic, as in most clinical trials, we can extend our findings to clinical trials and show that

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side effects can have a major effect on blinding success, i.e. the participant's expectations of having received treatment or having received no treatment. This is consistent with previous findings that show unblinding by side effects as an important problem in clinical trials.[33-35] Here, we also show that these treatment expectations then lead to differences in placebo effect. Our data support that side effects can differentially influence placebo effects when side effect occurrence differs between experimental conditions, as is often the case in clinical trials.[36,37] The validity of clinical trials depends on the additivity of non-specific factors (i.e. placebo effect). Our data show that this is not the case if side effects differ between treatment and control arm and therefore question the validity of the additivity assumption in these cases. As a consequence, clinical trials could overestimate the effect of a drug if side effects increase placebo effects in the treatment condition. Active placebos in the control arm[9] or other innovative research designs[38] could counteract this confound and improve clinical trial validity. Taken together, our data show the significant influence of side effects on treatment expectations and placebo effects. Taking these effects into account could improve clinical practice as well as clinical trials.

Data availability

The data that support the findings of this study are available in G-Node at https://doi.org/10.12751/g-node.5rvqjk.

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Competing interests

The authors report no competing interests.

Supplementary material

Supplementary material is available at *Brain* online.

References

1. Petrie KJ, Rief W. Psychobiological mechanisms of placebo and nocebo effects: Pathways to improve treatments and reduce side effects. *Annu Rev Psychol*. 2019;70:599-625.
2. Colloca L, Miller FG. The nocebo effect and its relevance for clinical practice. *Psychosom Med*. 2011;73:598-603.
3. Berna C, Kirsch I, Zion SR, et al. Side effects can enhance treatment response through expectancy effects: An experimental analgesic randomized controlled trial. *Pain*. 2017;158:1014-1020.
4. Benedetti F. Placebo and the new physiology of the doctor-patient relationship. *Physiol Rev*. 2013;93:1207-1246.
5. Kaptchuk TJ, Miller FG. Placebo effects in medicine. *N Engl J Med*. 2015;373:8-9.
6. Sanders AE, Slade GD, Fillingim RB, Ohrbach R, Arbes SJ Jr, Tchivileva IE. Effect of treatment expectation on placebo response and analgesic efficacy: A secondary aim in a randomized clinical trial. *JAMA Netw Open*. 2020;3:e202907.
7. Rief W, Glombiewski JA. The hidden effects of blinded, placebo-controlled randomized trials: An experimental investigation. *Pain*. 2012;153:2473-2477.
8. Linde K, Atmann O, Meissner K, et al. How often do general practitioners use placebos and non-specific interventions? Systematic review and meta-analysis of surveys. *PLoS One*. 2018;13:e0202211.
9. Jensen JS, Bielefeldt AØ, Hróbjartsson A. Active placebo control groups of pharmacological interventions were rarely used but merited serious consideration: A methodological overview. *J Clin Epidemiol*. 2017;87:35-46.
10. Bingel U, Lorenz J, Schoell E, Weiller C, Büchel C. Mechanisms of placebo analgesia: rACC recruitment of a subcortical antinociceptive network. *Pain*. 2006;120(1–2):8-15.
11. Eippert F, Bingel U, Schoell ED, et al. Activation of the opioidergic descending pain control system underlies placebo analgesia. *Neuron*. 2009;63:533-543.
12. Montgomery GH, Kirsch I. Classical conditioning and the placebo effect. *Pain*. 1997;72(1–2):107-113.
13. Horne R, Weinman J, Hankins M. The beliefs about medicines questionnaire: The development and evaluation of a new method for assessing the cognitive representation of medication. *Psychol Health*. 1999;14:1-24.
14. Clement LM, Bradley-Garcia M. A step-by-step tutorial for performing a moderated mediation analysis using PROCESS. *Quant Methods Psychol*. 2022;18:258-271.
15. Atlas LY, Wager TD. A meta-analysis of brain mechanisms of placebo analgesia: Consistent findings and unanswered questions. In: Benedetti F, Enck P, Frisaldi E, Schedlowski M, eds. *Placebo*. Springer; 2014:37-69.
16. Linnman C, Moulton EA, Barmettler G, Becerra L, Borsook D. Neuroimaging of the periaqueductal gray: State of the field. *Neuroimage*. 2012;60:505-522.
17. Friston KJ, Buechel C, Fink GR, Morris J, Rolls E, Dolan RJ. Psychophysiological and modulatory interactions in neuroimaging. *Neuroimage*. 1997;6:218-229.
18. Wager TD, Atlas LY. The neuroscience of placebo effects: Connecting context, learning and health. *Nat Rev Neurosci*. 2015;16:403-418.
19. Bingel U, Tracey I. Imaging CNS modulation of pain in humans. *Physiology*. 2008;23:371-380.
20. Wager TD, Rilling JK, Smith EE, et al. Placebo-Induced changes in fMRI in the anticipation and experience of pain. *Science*. 2004;303:1162-1167.
21. Peciña M, Azhar H, Love TM, et al. Personality trait predictors of placebo analgesia and neurobiological correlates. *Neuropsychopharmacology*. 2013;38:639-646.
22. Berna C, Leknes S, Holmes EA, Edwards RR, Goodwin GM, Tracey I. Induction of depressed mood disrupts emotion regulation neurocircuitry and enhances pain unpleasantness. *Biol Psychiatry*. 2010;67:1083-1090.
23. Zunhammer M, Spisák T, Wager TD, Bingel U. Meta-analysis of neural systems underlying placebo analgesia from individual participant fMRI data. *Nat Commun*. 2021;12:1391.

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24. Petersen KK, McPhee ME, Hoegh MS, Graven-Nielsen T. Assessment of conditioned pain modulation in healthy participants and patients with chronic pain: Manifestations and implications for pain progression. *Curr Opin Support Palliat Care*. 2019;13:99-106.
25. Granot M, Weissman-Fogel I, Crispel Y, et al. Determinants of endogenous analgesia magnitude in a diffuse noxious inhibitory control (DNIC) paradigm: Do conditioning stimulus painfulness, gender and personality variables matter? *Pain*. 2008;136:142-149.
26. Colloca L, Barsky AJ. Placebo and nocebo effects. *N Engl J Med*. 2020;382:554-561.
27. Kaptchuk TJ, Miller FG. Open label placebo: Can honestly prescribed placebos evoke meaningful therapeutic benefits? *BMJ*. 2018;363:k3889.
28. Leibowitz KA, Howe LC, Crum AJ. Changing mindsets about side effects. *BMJ Open*. 2021;11:e040134.
29. Evers AWM, Colloca L, Blease C, et al. Implications of placebo and nocebo effects for clinical practice: Expert consensus. *Psychother Psychosom*. 2018;87:204-210.
30. Manaï M, van Middendorp H, Veldhuijzen DS, Huizinga TWJ, Evers AWM. How to prevent, minimize, or extinguish nocebo effects in pain: A narrative review on mechanisms, predictors, and interventions. *Pain Rep*. 2019;4:e699.
31. Bingel U. Placebo 2.0: The impact of expectations on analgesic treatment outcome. *Pain*. 2020;161:S48-S56.
32. Sibbald B, Roland M. Understanding controlled trials: Why are randomised controlled trials important? *BMJ*. 1998;316:201.
33. Turner JA, Jensen MP, Warms CA, Cardenas DD. Blinding effectiveness and association of pretreatment expectations with pain improvement in a double-blind randomized controlled trial. *Pain*. 2002;99:91-99.
34. Basoglu M, Marks I, Livanou M, Swinson R. Double-blindness procedures, rater blindness, and ratings of outcome: Observations from a controlled trial. *Arch Gen Psychiatry*. 1997;54:744-748.
35. Moncrieff J. A comparison of antidepressant trials using active and inert placebos. *Int J Methods Psychiatr Res*. 2003;12:117-127.
36. Bell RF, Eccleston C, Kalso EA. Ketamine as an adjuvant to opioids for cancer pain. *Cochrane Database Syst Rev*. 2017;6:CD003351.
37. Els C, Jackson TD, Kunyk D, et al. Adverse events associated with medium- and long-term use of opioids for chronic non-cancer pain: An overview of Cochrane Reviews. *Cochrane Database Syst Rev*. 2017;2017:CD012509.
38. Colagiuri B. Participant expectancies in double-blind randomized placebo-controlled trials: Potential limitations to trial validity. *Clin Trials*. 2010;7:246-255.

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